

Problem Statement – ABC Campaign Analysis

Context:

- ABC Supermarket is planning for the year-end sale - they want to launch a new offer i.e. gold membership for only \$499 that is of \$999 on normal days (that gives 20% discount on all purchases) only for existing customers, for that they need to do a campaign through phone calls - the best way to reduce the cost of the campaign is to make a predictive model to classify customers who might purchase the offer, using the data they gathered during last year's campaign.
- We will build a model for classifying whether customers will reply with a positive response or not.

Problem:

The data-set aims to answer the following key questions:

- What are the different factors which affect the target variable?
- How can we improve model performance using hyperparameter tuning and prevent data leakage using pipelines while building a model to predict the response of a customer?
- What business recommendations can we give based on the analysis and how do you recommend them to use the model?
- Based on the model results, how can we measure the value the model brings?

Data Dictionary

- The data contains characteristics of the people
 - Response (target) - 1 if customer accepted the offer in the last campaign, 0 otherwise
 - Complain - 1 if a customer complained in the last 2 years
 - DtCustomer - date of customer's enrolment with the company
 - Education - customer's level of education
 - Marital - customer's marital status
 - Kidhome - number of small children in customer's household
 - Teenhome - number of teenagers in customer's household
 - Income - customer's yearly household income
 - MntFishProducts - the amount spent on fish products in the last 2 years
 - MntMeatProducts - the amount spent on meat products in the last 2 years
 - MntFruits - the amount spent on fruits products in the last 2 years

- MntSweetProducts - amount spent on sweet products in the last 2 years
- MntWines - the amount spent on wine products in the last 2 years
- MntGoldProds - the amount spent on gold products in the last 2 years
- NumDealsPurchases - number of purchases made with discount
- NumCatalogPurchases - number of purchases made using catalog
- NumStorePurchases - number of purchases made directly in stores
- NumWebPurchases - number of purchases made through the company's web site
- NumWebVisitsMonth - number of visits to company's web site in the last month
- Recency - number of days since the last purchase
- ID - unique customer-id
- Year_Birth - customer's year of birth

Solution Expectation:

Please prepare a notebook (Jupyter/Colab/Databricks) with clear comments and explanation and a slide deck that covers the following -

- Import Libraries and Load Dataset
- Overview of data
- Data Visualization and EDA
- Data preparation
- Train various models – with comments on why certain models are not doing well
- Tune models using grid/random search or using statistical analysis
- Provide recommendations to the business
- Conclusion

Final Deliverables:

- 1) Kindly zip all your analysis (including notebook & slides) with your full name and date.
- 2) Please also add the dataset in the zip file to make it a consumable notebook